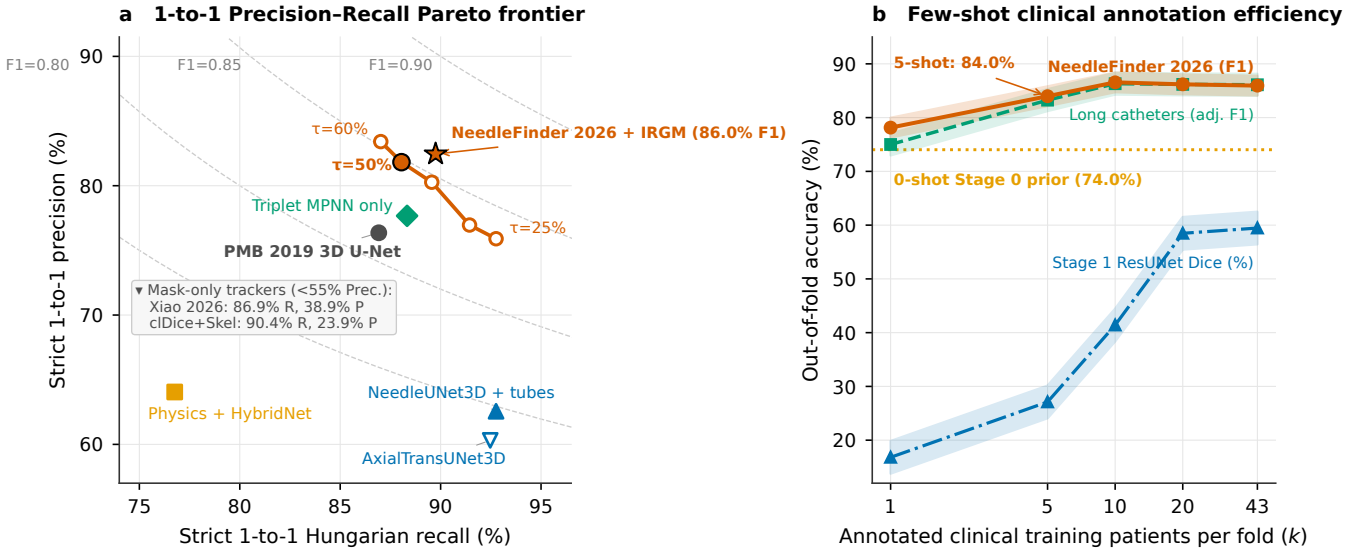
a 1-to-1 Precision–Recall Pareto frontier
F1=0.80
F1=0.85
F1=0.90
τ=60%
τ=50%
τ=25%
NeedleFinder 2026 + IRGM (86.0% F1)
Triplet MPNN only
PMB 2019 3D U-Net
▾ Mask-only trackers (<55% Prec.):
Xiao 2026: 86.9% R, 38.9% P
clDice+Skel: 90.4% R, 23.9% P
Physics + HybridNet
NeedleUNet3D + tubes
AxialTransUNet3D
Strict 1-to-1 precision (%)
Strict 1-to-1 Hungarian recall (%)
90
80
70
60
75
80
85
90
95
b Few-shot clinical annotation efficiency
5-shot: 84.0%
NeedleFinder 2026 (F1)
Long catheters (adj. F1)
0-shot Stage 0 prior (74.0%)
Stage 1 ResUNet Dice (%)
Out-of-fold accuracy (%)
Annotated clinical training patients per fold (k)
90
80
70
60
50
40
30
20
1
5
10
20
43